

z -statistic:

$$\begin{aligned} z &= \frac{p_{\text{StrongREJECT}}^{(\text{o1})} - p_{\text{StrongREJECT}}^{(\text{Sonnet})}}{SE} \\ &= \frac{0.92 - 0.90}{0.0229} \\ &\approx 0.87. \end{aligned}$$

This yields a p -value: ≈ 0.38 , which is not significant. Meanwhile, results from the automated classifier suggested that the difference is highly significant at $p \approx 0.004$!

The case study illustrates that significant problems may arise from relying on automated judges alone, as—in addition to unavoidable noise due to classification errors—they may have systematic biases that distort model comparisons. Here, we saw that Claude’s true robustness was underestimated by 11.1%, while o1’s was only underestimated by 4.5%, which directly affects the conclusions we can confidently draw from the experiments. We thus reiterate our suggestion to verify judge performance via human evaluation when proposing a new attack, defense, or model to ensure comparable results.

A.6. Case Study: Comparing the StrongREJECT Gemma 2B and HarmBench/CAIS Llama 2 13B judges.

We compare the results provided by StrongREJECT’s (Souly et al., 2024) Gemma 2B judge model and the HarmBench (Mazeika et al., 2024) Llama 2 13B finetuned classifier.

To do so, we run a suite of 10 different attack algorithms against a zoo of 25 models using all 300 non-copyright prompts in HarmBench’s dataset. All attacks are evaluated in the “many-trial” setting, regardless of the setting in which they were originally introduced. This means all intermediate prompt candidates are rolled-out and evaluated. In total, we compare results for 5,269,564 generations. We acknowledge that due to edge-case problems with models, tokenizers, and attack implementations, and due to time constraints, not all attack runs were successfully completed. Nonetheless, as we only compare the *difference*, we can nonetheless make valid comparisons between the two judge models. Our results, shown in Figure 8, highlight substantial model- and attack-specific discrepancies between these two commonly used judges. While StrongREJECT’s consistently reports lower ASRs (as expected, due to its more strict definition of what constitutes a successful jailbreak), we find very clear *differences in how much the two judges differ*. These inconsistencies can lead to differences in the scoring and even ranking of models, defenses, and attacks, complicating comparisons.

Ultimately, we would like to compare these classifier-based judgments to human-provided ground truth, but this was not feasible within the scope of this work. Nonetheless, if these two commonly used classifiers disagree, at least one (and likely both) must also disagree with human ground truth, underscoring the limitations of automated evaluation.

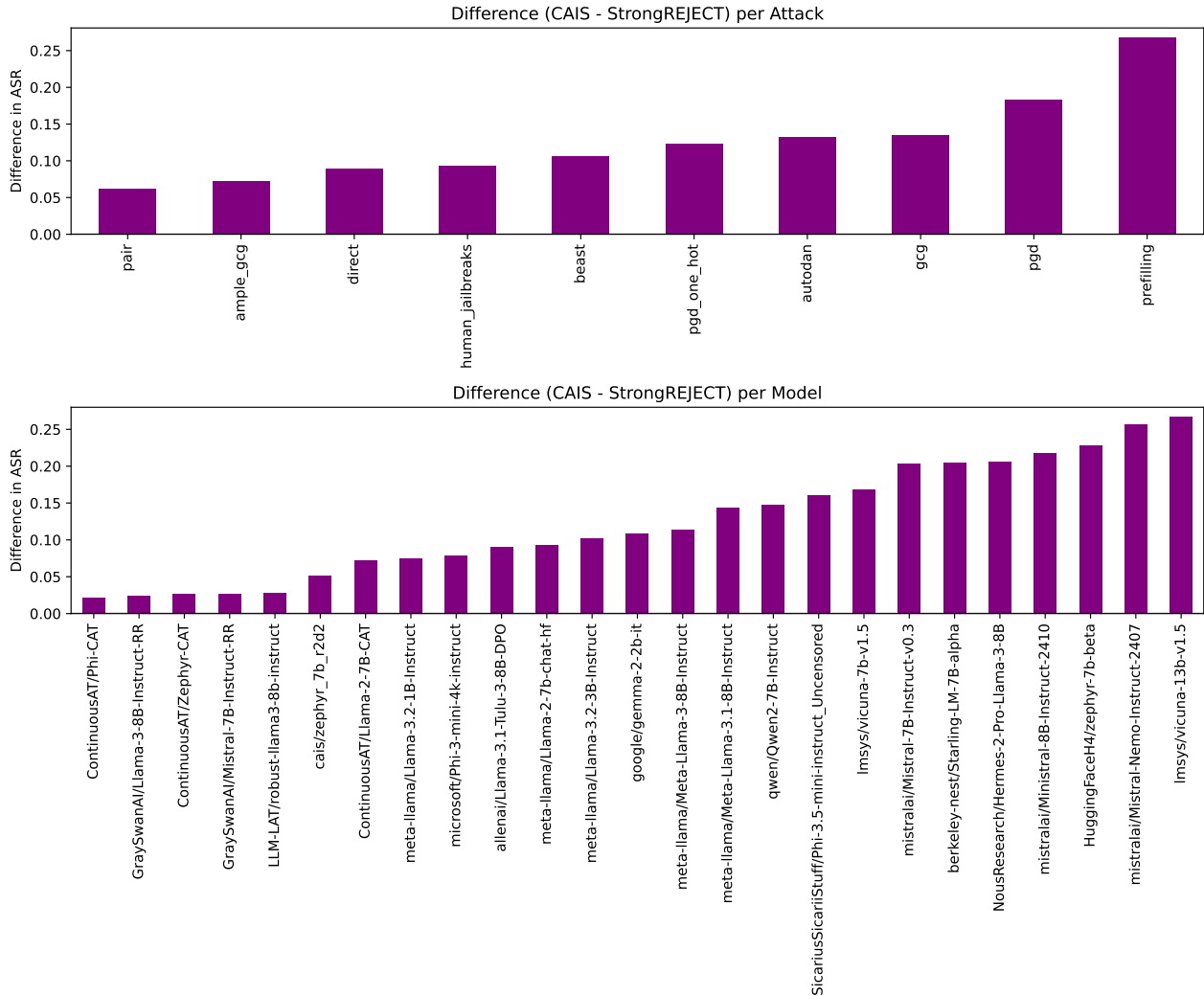


Figure 8: Comparing the ASR of two judge models. We consider an attack successful if StrongReject gives a score higher than 0.5, and if HarmBench’s classifier outputs “Yes” or “yes” as the most likely token.

We briefly describe the hyperparameters we used below. All attacks implementations were adapted from the official GitHub repositories of the original authors where available. Otherwise, we ported a HarmBench implementation to our pipeline. In some cases, we conferred with authors directly to gain access to reference implementations and ensure correctness. For all attacks, we sample and judge a single greedy generation per prompt-candidate.

- PAIR (Chao et al., 2023): We use lmsys/vicuna-13b-v1.5 as the attacker model and generate up to 512 tokens per attack prompt. We sample with temperature 1 and top-p of 0.9, setting N_{streams} to 5 and $N_{\text{iterations}}$ to 5. During the attack, the victim model generates up to 256 tokens using greedy generation.
- AmpleGCG (Liao & Sun, 2024): We use osunlp/AmpleGCG-llama2-sourced-llama2-7b-chat to generate 200 attack suffixes with diversity penalty 1.
- Direct: We simply use the harmful prompt without any modification and sample a greedy generation.
- HumanJailbreaks: We use the 114 human-designed jailbreak templates in HarmBench (Mazeika et al., 2024) to prompt the model.
- BEAST (Sadasivan et al., 2024): We use $k_1 = k_2 = 15$ and set the temperature to 1 to sample $N = 40$ suffix tokens.
- PGD (Schwinn et al., 2024): We initialize the attack using the suffix “x x x x x x x x x x x x x x x x x x” as it tokenizes to exactly 20 tokens for all tested models, and run signed gradient descent optimization for 100 steps. We use a learning rate of $\alpha = 0.001$ and constrain the optimization to an L_2 ball with radius 1 around the initialization for each token. In the one-hot version, instead of optimizing the input embeddings directly in latent space, we instead optimize over the $\{0, 1\} \rightarrow [0, 1]$ relaxed one-hot input vectors. In addition, we found it necessary to add a momentum term with weight 0.5 to speed up the optimization.
- AutoDAN (Liu et al., 2023a): We use 100 steps and initialize using the 128 seed prompts from HarmBench’s implementation. We use the victim model itself as mutator model and set $N_{\text{elites}} = 0.05$, crossover = 0.5, $N_{\text{points}} = 5$, and $P_{\text{mutation}} = 0.01$.
- GCG (Zou et al., 2023b): We use our modified version of nanoGCG with a corrected token filtering algorithm to remove special tokens & only allow ASCII-representable characters. We set $N_{\text{steps}} = 250$, use a batch size of 512, Top-K = 256, and initialize using the string “x x x x x x x x x x x x x x x x x x” as it tokenizes to exactly 20 tokens for all tested models.
- Prefilling: In this attack, we use the unmodified harmful prompt and pre-fill the beginning of the model’s response using the affirmative target sequence from the dataset.

A.7. Reproducibility

We will release code to exactly reproduce all⁴ plots and tables upon publication. All experiments were conducted using greedy generation, bfloat16 precision, and HuggingFace Transformers version 4.47.1 unless otherwise noted. To judge generations we use the CAIS/HarmBench Llama 2 13B classifier and the finetuned Gemma 2B from StrongREJECT. For the HarmBench classifier, a generation is deemed harmful if the most likely output token is either “yes” or “Yes” and for StrongREJECT, we threshold the score at 0.5. We do not truncate model generations (i.e. we judge the full model response up to an end-of-sentence or end-of-turn token).

To obtain the results in Figures 4 & 5 and Table 1, we used a batch size of 512 and 250 steps of GCG with top-k set to 256. We initialized the suffix with the string “x x x x x x x x x x x x x x x x x x”.

⁴Due to the large scale of the experiments and the associated rare occurrence of edge-cases and other non-deterministic problems, it may not be possible to exactly reproduce Figure 8 from scratch.